



Modelling Financial Markets

Chapter 19: Review and Project Presentations



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ ASE: Academia de Studii Economice din București (the Bucharest University of Economic Studies)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CV: Cross-Validation
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DM: Diebold–Mariano (test)
- ▣ DQ: Dynamic Quantile (test)
- ▣ ES: Expected Shortfall
- ▣ ETF: Exchange-Traded Fund
- ▣ EVT: Extreme Value Theory
- ▣ FDR: False Discovery Rate
- ▣ FHS: Filtered Historical Simulation
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FWER: Family-Wise Error Rate
- ▣ FX: Foreign eXchange
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/2)

- ▣ GARCH-EVT: GARCH filter combined with Extreme Value Theory for the tail
- ▣ GBM: Geometric Brownian Motion
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ GRS: Gibbons–Ross–Shanken (test)
- ▣ GSADF: Generalised Sup Augmented Dickey–Fuller (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HS: Historical Simulation
- ▣ IGARCH: Integrated GARCH
- ▣ IQR: InterQuartile Range
- ▣ LRMEs: Long-Run Marginal Expected Shortfall
- ▣ MCS: Model Confidence Set
- ▣ ML: Machine Learning
- ▣ MSE: Mean Squared Error
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ QMLE: Quasi-Maximum Likelihood Estimator
- ▣ SADF: Sup Augmented Dickey–Fuller (test)
- ▣ SE: Standard Error
- ▣ SML: Security Market Line
- ▣ SRISK: Systemic RISK (capital shortfall measure)
- ▣ US: United States
- ▣ VIX: CBOE Volatility Index
- ▣ VOLARE: VOLatility Archive for Realized Estimates (database)

Outline

Introduction

Review I: Markets, Returns and Pricing

Review II: Volatility, Dependence and Risk

Review III: High Frequency, Continuous Time and Options

Review IV: Learning, Text and New Markets

Common Pitfalls

Integrated Case Study: BET

The Team Project

Using AI in Your Project

Project Seeds

Case Study: Nonstandard Errors

AI for Scientific Discovery

Conclusions

References

What This Chapter Does

Purpose

- Review Chapters 0–18 as one connected story
- Collect the pitfalls that returned in every chapter
- Run one integrated case study, from raw prices to a backtested value-at-risk (VaR) at 1%
- Prepare the team project: its four steps, the question, the methods and the AI policy

Assessment

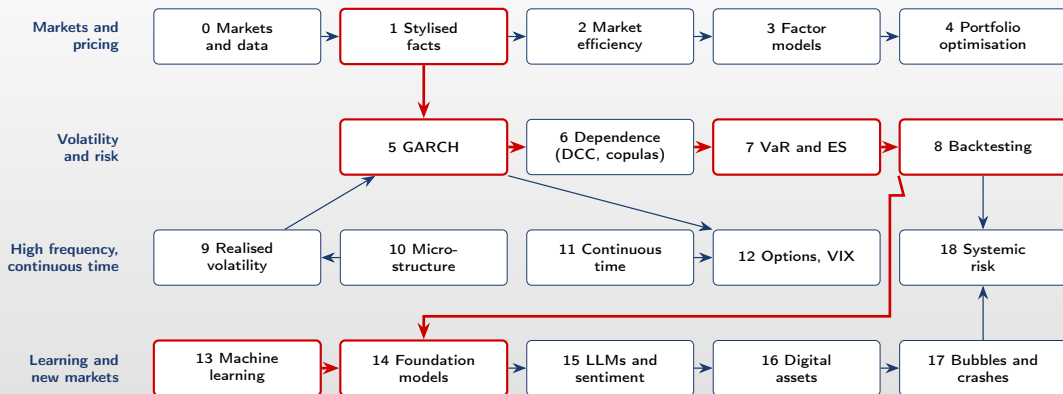
- Team project (GitHub, presentation, oral defence): **70%**
- Quizzes and seminar activity: **20%**
- Attendance: **10%**; there is no final exam



Bucharest University of Economic Studies (ASE), Piața Romană

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The Course in One Picture



- Red: the main thread of the course, from stylised facts to risk forecasts that we can test
- Blue arrows: where one chapter supplies the input or the benchmark of another

Reading the Course Map

- Chapter 1 sets the targets: heavy tails, volatility clustering, almost no linear predictability
 - ▶ Every later model is judged by whether it reproduces these facts
- Chapters 5–8 turn the facts into a risk number and a test of that number
 - ▶ GARCH gives σ_{t+1} ; VaR and ES give the loss; backtesting says whether to trust it
- Chapters 9–12 measure volatility better (intraday data, option prices) and in continuous time
- Chapters 13–15 bring flexible models, always evaluated with the tools of Chapter 8
- Chapters 16–18 apply the same toolkit to crypto, bubbles and the whole financial system
- Special Chapter 20: forecasting realised volatility with path signatures on the VOLARE data

Five Questions for Every Model

Before estimating

- **Data:** which series, which calendar, which adjustments (dividends, splits, holidays)?
- **Model:** which stylised fact does it capture, and which does it ignore?
- **Estimation:** on which window, with which information at time t ?

After estimating

- **Evaluation:** out of sample, with a loss that matches the target (QLIKE, FZ0) and the right diagnostic tests (e.g. Kupiec)?
- **Uncertainty:** robust standard errors, bootstrap, a test against a simple benchmark?
- If one answer is missing, the result is not yet a finding

Chapter 0: Financial Markets in 2026

Key idea

- ▣ Markets combine exchanges, over-the-counter dealers, exchange-traded funds (ETFs), algorithms and blockchains
- ▣ Data quality is a modelling decision: source, adjustment and calendar change the result

Key inference

- ▣ For i.i.d. log returns, annualised by the number of periods a year: $se(\hat{\mu}_{\text{ann}}) = \sigma_{\text{ann}} / \sqrt{Y}$ over Y years, whatever the frequency; $se(\hat{\sigma}) \approx \sigma \sqrt{(K-1)/(4T)}$ over T days, K the population kurtosis (finite fourth moment needed); serial dependence calls for long-run variances
- ▣ SPY 2015–2026: $se(\hat{\mu}_{\text{ann}}) \approx 5.1\%$ for a mean of 13.7%: means are barely estimable, volatilities are

Result from the data

- ▣ 2015–2026: SPY (S&P 500 ETF, dividends reinvested) 13.7% a year with 17.6% volatility; Bitcoin 60.6% a year with 66.7%
- ▣ BET, July 2007 to February 2009: a drawdown of −82.5%

Leads to

- ▣ Returns, not prices, are the object of study (Chapter 1)

Chapter 1: Stylised Facts and Returns

Key idea

- ▣ Returns are uncorrelated but not independent: their magnitude is predictable, but their sign is largely unpredictable
- ▣ Heavy tails and volatility clustering in every market [Cont \(2001\)](#)

Key inference

- ▣ Ljung–Box $Q(m) \sim \chi_m^2$ needs i.i.d. returns; under volatility clustering it over-rejects: use the robust Q^* with $\hat{\rho}_k^2$ weighted by $T\hat{\rho}_k^2/\hat{\tau}_k$ [Lobato, Nankervis & Savin \(2001\)](#)
- ▣ Tails: Hill $\hat{\alpha}$ with $se \approx \hat{\alpha}/\sqrt{k}$ [Hill \(1975\)](#); if $\alpha < 4$ the sample kurtosis is not consistent and its bootstrap fails [Athreya \(1987\)](#)

Result from the data

- ▣ BET, Hill index of losses (top $k = 153$): $\hat{\alpha} = 2.39$, block-bootstrap 95% CI [2.12, 3.03]: the fourth moment is likely infinite
- ▣ Days beyond 5σ are more than 1000 times more frequent than the Normal distribution predicts

Leads to

- ▣ Volatility models (Chapter 5) and tail risk measures (Chapter 7)

Chapter 2: Market Efficiency

Key idea

- Efficiency: prices fully reflect the information set; a test of abnormal-return predictability is a joint test of efficiency and the expected-return model [Fama \(1970\)](#)
- Efficiency varies over time (adaptive markets); tests need robust inference

Key inference

- Heteroskedasticity-robust variance ratio $z^*(q) \Rightarrow N(0, 1)$ under a martingale difference null [Lo & MacKinlay \(1988\)](#)
- Several horizons q at once: the joint test $\max_q |z^*(q)|$ with studentised maximum modulus critical values [Chow & Denning \(1993\)](#); rolling windows need a multiplicity correction too

Result from the data

- Share of rolling windows that reject the random walk: S&P 500 6.4%, BET 22.4%, Bitcoin 0.0%
- Statistical rejection is not exploitable profit

Leads to

- Anomalies and factors (Chapter 3); the multiple-testing problem

Chapter 3: Factor Models and Asset Pricing

Key idea

- Expected returns should reward only exposure to systematic factors
- Hundreds of published factors require higher hurdles Harvey, Liu & Zhu (2016)

Key inference

- $$GRS = \frac{T-N-K}{N} \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{\mu}_f^\top \hat{\Omega}^{-1} \hat{\mu}_f} \sim F(N, T-N-K) \text{ only}$$
 under i.i.d. Normal errors Gibbons, Ross & Shanken (1989)
- Fama–MacBeth s.e. ignore that betas are estimated: with $c = \hat{\gamma}^\top \hat{\Sigma}_f^{-1} \hat{\gamma}$, Shanken's covariance is $(1+c)(\hat{V}_{FM} - \hat{\Sigma}_f/T) + \hat{\Sigma}_f/T$ for estimated factor loadings Shanken (1992)

Result from the data

- 25 size and book-to-market portfolios: SML slope -4.4% , $GRS = 4.18$: the CAPM is rejected
- Sector ETFs: the first principal component (the market) explains 66% of the variance

Leads to

- Portfolio construction (Chapter 4); factor models inside machine learning (Chapter 13)

Chapter 4: Portfolio Optimisation

Key idea

- Markowitz asks the right question, but estimation error dominates the answer [Markowitz \(1952\)](#)
- Shrinkage, constraints and risk-based rules stabilise the weights; $1/N$ is a hard benchmark [DeMiguel, Garlappi & Uppal \(2009\)](#)

Key inference

- $\text{se}(\widehat{SR}) \approx \sqrt{(1 + \widehat{SR}^2/2)/T}$ for i.i.d. Normal returns, per period [Lo \(2002\)](#); serial correlation changes the annualisation factor
- Is portfolio A better than $1/N$? Test $SR_A - SR_B = 0$ with HAC or studentised bootstrap inference [Ledoit & Wolf \(2008\)](#), not by comparing two point estimates

Result from the data

- 60-month samples: median in-sample Sharpe 1.57, median true Sharpe 0.30
- 96% of the estimated tangency portfolios are truly worse than $1/N$

Leads to

- Risk budgets need volatility and dependence forecasts (Chapters 5–6)

Recap: Markets, Returns and Pricing

- ▣ Clean data first: adjusted prices for ETFs and stocks, own calendar for each series, join on prices before returns
- ▣ Stylised facts are the benchmark for every model that follows
- ▣ Efficiency and factor tests need robust errors and multiple-testing corrections
- ▣ Optimisation amplifies estimation error: evaluate out of sample, against $1/N$

Chapter 5: Conditional Volatility: GARCH

Key idea

- Volatility clusters, so today's shock predicts tomorrow's variance [Engle \(1982\)](#), [Bollerslev \(1986\)](#)
- Persistence close to 1; asymmetry in equities; heavy-tailed innovations

Key inference

- QMLE: $\sqrt{T}(\hat{\theta} - \theta) \Rightarrow N(0, A^{-1}BA^{-1})$ (sandwich); Hessian-only s.e. are wrong for non-Normal innovations [Bollerslev & Wooldridge \(1992\)](#)
- Near $\alpha + \beta = 1$ the parameter is close to the boundary of the covariance-stationary region: Wald intervals are unreliable; use profile likelihood or bootstrap restricted to the parameter space [Andrews \(1999\)](#)

Result from the data

- S&P 500, 2000–2026: $\alpha + \beta = 0.9947$, a half-life of 131 days
- BET: profile-likelihood 95% CI for $\alpha + \beta$ is $[0.975, 1]$ (IGARCH not rejected): the half-life is at least 27 days, with no finite upper bound

Leads to

- $\hat{\sigma}_{t+1}$ is the input of filtered VaR and ES (Chapter 7)

Chapter 6: Multivariate Volatility and Dependence

Key idea

- ▣ Correlations move with the macro regime; diversification is not guaranteed
- ▣ Copulas separate the margins from dependence; tail dependence drives joint losses

Key inference

- ▣ DCC is estimated in two steps: second-step s.e. must account for the first-step volatilities [Engle \(2002\)](#)
- ▣ Crisis vs calm correlation: ρ rises mechanically with volatility; test on $\rho^* = \rho / \sqrt{1 + \delta(1 - \rho^2)}$, δ the relative rise in variance [Forbes & Rigobon \(2002\)](#)

Result from the data

- ▣ SPY–TLT (long US Treasury bond ETF): ρ from -0.40 to 0.11 after 2022, so a 60/40 portfolio went from 10.4% to 13.2% volatility
- ▣ Bitcoin–SPY DCC correlation: 0.04 before 2020, 0.31 since

Leads to

- ▣ Portfolio VaR and Euler contributions (Chapter 7); systemic risk (Chapter 18)

Chapter 7: Value-at-Risk and Expected Shortfall

Key idea

- VaR is a quantile of the loss, ES the average loss beyond it; ES is coherent
- Filtered historical simulation and conditional EVT react to volatility in days McNeil & Frey (2000)

Key inference

- Historical VaR, i.i.d. returns with a continuous density $f(q_\alpha) > 0$: $\sqrt{T}(\hat{q}_\alpha - q_\alpha) \Rightarrow N(0, \alpha(1 - \alpha)/f(q_\alpha)^2)$; in the tail f is small, so the estimate is noisy; under dependence $\alpha(1 - \alpha)$ becomes the long-run variance of the hits $\mathbb{1}\{r_t \leq q_\alpha\}$
- ES averages the αT largest losses: with αT of a few dozen and a tail index near 3, its variance is dominated by one or two days

Result from the data

- S&P 500, daily: historical $\text{VaR}_{1\%} = 3.45\%$, $\text{ES}_{1\%} = 5.01\%$
- The Normal distribution gives only 2.80%: heavy tails matter

Leads to

- A risk forecast is useful only if it survives a backtest (Chapter 8)

Chapter 8: Backtesting and Evaluating Risk Forecasts

Key idea

- Test the number and the timing of breaches: coverage and independence [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#)
- Rank (VaR, ES) forecasts with a consistent score (FZ0) and Diebold–Mariano tests [Fissler & Ziegel \(2016\)](#)

Key inference

- Kupiec $LR_{uc} \sim \chi_1^2$, Christoffersen $LR_{ind} \sim \chi_1^2$; the DQ test regresses $I_t - \alpha$ on lagged hits and Var_t , $DQ \sim \chi_k^2$ [Engle & Manganelli \(2004\)](#)
- Power is low: at $T = 250$ the 5% Kupiec test detects a true rate of 2% only 24% of the time; compare models with conditional predictive ability tests [Giacomini & White \(2006\)](#)

Result from the data

- S&P 500, VaR 1%, 2007–2026: Normal 2.70%, HS 1.67%, FHS 1.23%, GARCH-EVT 1.13% breaches
- A GARCH filter plus a flexible tail wins in all four markets

Leads to

- Forecast-comparison tests (DM, MCS) with a suitable loss also judge realised-volatility forecasts; coverage tests judge quantile forecasts, including machine-learning ones (Chapters 9, 13, 14)

Recap: Volatility, Dependence and Risk

- ▣ GARCH turns volatility clustering into one-day-ahead forecasts
- ▣ Dependence is dynamic and strongest in the tails; crisis correlations need a volatility adjustment
- ▣ VaR and ES must combine a volatility filter with a heavy-tailed distribution
- ▣ A risk model is accepted only after tests of coverage, independence and ES

Chapter 9: Realised Volatility

Key idea

- Intraday prices measure daily variance (RV , realised variance); the log-HAR (heterogeneous autoregressive) model captures its long memory [Corsi \(2009\)](#)
- Returns divided by realised volatility have a distribution close to the Normal distribution

Key inference

- HAR errors are heteroskedastic and, for multi-day targets that overlap, serially correlated: use HAC s.e.; forecasts are compared with DM tests on a robust loss [Patton \(2011\)](#)
- Microstructure noise biases RV upwards at high frequency: two-scale estimators remove the bias [Zhang, Mykland & Aït-Sahalia \(2005\)](#)

Result from the data

- SPY: kurtosis 15.73 for open-to-close returns, 2.65 after dividing by $\sqrt{RV_t}$
- log-HAR beats GARCH(1,1)- t in QLIKE by 10% for SPY ($p = 0.021$) and 22% for Bitcoin

Leads to

- Better volatility inputs for VaR; microstructure explains the noise (Chapter 10)

Chapter 10: Market Microstructure

Key idea

- Prices form in limit order books; the spread pays for processing, inventory and adverse selection
- Liquidity follows the clock and the VIX, and disappears in stress

Key inference

- The Roll spread needs $\text{Cov}(\Delta p_t, \Delta p_{t-1}) < 0$; when the sample covariance is positive the estimator is undefined, so report how often
- An illiquidity premium is a cross-sectional claim: Fama–MacBeth with Newey–West s.e. and the Shanken correction [Shanken \(1992\)](#)

Result from the data

- Amihud ratio (bp per USD 1 million): SPY 0.0016, Banca Transilvania 24.7, Transelectrica 1024.0
- SPY: the opening 5-minute bar moves 2.06 times as much as a midday bar

Leads to

- Liquidity risk in VaR and in the project data choices

Chapter 11: Continuous-Time Models

Key idea

- Brownian motion is the limit of random walks; Itô's lemma solves the price equations
- Jumps explain the tails, stochastic volatility the clustering

Key inference

- Normality of GBM log returns: Jarque–Bera $\sim \chi^2_2$ rejects everywhere; jumps are tested with bipower variation, $(RV - BV)$ standardised $\Rightarrow N(0, 1)$ Barndorff-Nielsen & Shephard (2006)
- The test is daily: over many days, control the number of false jump days

Result from the data

- S&P 500 against geometric Brownian motion: 62 days beyond ± 4 standard deviations, against 0.59 expected
- Hill index of S&P 500 losses: 2.87 [2.07, 3.67] (Chapter 16): the fourth moment is likely infinite, unlike the Normal log returns of GBM

Leads to

- Option pricing needs only σ , not μ (Chapter 12)

Chapter 12: Options and the Volatility Surface

Key idea

- Black–Scholes prices options by hedging; implied volatility is the market's view of σ [Black & Scholes \(1973\)](#)
- Smile and skew encode the fear of crashes

Key inference

- VRP_t uses the next 21 days of realised variance: consecutive values overlap, so the mean needs Newey–West s.e. with at least 21 lags
- Implied volatility is a risk-neutral forecast: the premium, not a forecast error, explains most of the gap

Result from the data

- 1990–2026: mean VIX 19.4, mean realised volatility over the next month 15.3
- The variance risk premium is positive on 86% of days, but not in crashes

Leads to

- Implied volatility as a feature for machine learning and as a stress indicator (Chapters 13, 18)

Recap: High Frequency, Continuous Time and Options

- ▣ Intraday data measure volatility; log-HAR forecasts it better than daily GARCH
- ▣ Liquidity is a risk of its own and varies by orders of magnitude across markets
- ▣ Continuous-time models are tractable, but realistic ones need jumps and random volatility
- ▣ Option prices carry forward-looking information, including a variance risk premium

Chapter 13: Machine Learning in Finance

Key idea

- Flexible models, but little signal and much noise; the finance-specific part is the process [López de Prado \(2018\)](#)
- Purged cross-validation, meta-labelling and deflated Sharpe ratios [Bailey & López de Prado \(2014\)](#)

Key inference

- Selection among N strategies: the deflated Sharpe ratio tests $SR > E[\max_N \widehat{SR}]$ under the null of no skill [Bailey & López de Prado \(2014\)](#)
- Cross-validation on time series must purge overlapping labels and embargo the test fold, or its accuracy is biased upwards

Result from the data

- Shuffled K-fold reports 69.4% accuracy on pure noise: leakage
- S&P 500: the fractionally differenced series is stationary at $d^* = 0.25$; no model beats “always up”

Leads to

- Deep learning and foundation models, judged the same way (Chapter 14)

Chapter 14: Deep Learning and Time-Series Foundation Models

Key idea

- LSTMs (long short-term memory networks) and Transformers learn from sequences; foundation models forecast zero-shot
- Evaluate them with Chapter 8 tools, after the release of their weights

Key inference

- DM test on QLIKE differences with HAC variance: $t_{DM} \Rightarrow N(0, 1)$ for non-nested comparisons [Diebold & Mariano \(1995\)](#)
- With many candidate models, the model confidence set controls the family-wise error [Hansen, Lunde & Nason \(2011\)](#)

Result from the data

- SPY realised volatility, QLIKE: Chronos-2 0.231, log-HAR 0.234, GARCH- t 0.283; the first two are not significantly different
- Raw Chronos-2 VaR 1% on the S&P 500: 1.67% breaches

Leads to

- Text as data (Chapter 15)

Chapter 15: LLMs and Sentiment Analysis

Key idea

- From dictionaries to FinBERT and large language models (LLMs): text becomes a number
- Text embeddings enter forecasting models as features

Key inference

- Predictive regression $r_{t+1} = a + bS_t + \varepsilon_{t+1}$: a persistent S_t whose innovations correlate with returns biases $\hat{b}$ and the t -test [Stambaugh \(1999\)](#)
- Classifier accuracies on the same texts are paired: compare them with the exact McNemar test, not two separate intervals

Result from the data

- Qwen 7B vs 1.5B on financial headlines: +21.5 percentage points in accuracy, McNemar $p < 0.001$
- A headline-sentiment strategy: Sharpe +0.63 before and -1.48 after costs; break-even cost 1.5 bp per trade

Leads to

- Sentiment in crypto markets and bubbles (Chapters 16–17)

Chapter 16: Digital Assets and DeFi

Key idea

- ▣ Cryptocurrencies form a distinct asset class in the feature space of [Pele et al. \(2023\)](#), where a tail factor separates them; their Hill tail indices are not significantly lower than those of equities or gold (Chapter 16)
- ▣ Stablecoins, automated market makers and spot ETFs connect crypto to mainstream finance

Key inference

- ▣ Crypto factor models: GRS and Fama–MacBeth with the Shanken correction [Gibbons, Ross & Shanken \(1989\)](#), [Shanken \(1992\)](#); heavy tails ask for HAC or bootstrap versions
- ▣ Tails: Hill indices with intervals, not sample kurtosis (Chapter 1); a break in a correlation is tested with an unknown date [Andrews \(1993\)](#)

Result from the data

- ▣ Bitcoin–S&P 500 link: sup-Wald 20.2 (5% critical value 11.6), break estimated on 9 Mar 2020, not at the 2024 ETF launch
- ▣ Hill index of losses: Bitcoin 2.73 [2.12, 3.33], S&P 500 2.87 [2.07, 3.67]: statistically indistinguishable

Leads to

- ▣ Explosive price episodes (Chapter 17)

Chapter 17: Bubbles and Crashes

Key idea

- ▣ A bubble is a price path that grows faster than fundamentals can justify
- ▣ Right-tailed unit-root tests date the start and the end of explosive episodes

Key inference

- ▣ Under the unit-root null, SADF and GSADF converge to functionals of Brownian motion; critical values depend only on the minimum window r_0 Phillips, Shi & Yu (2015)
- ▣ Volatility shifts distort the size: use wild-bootstrap critical values; control the multiplicity of date-stamping

Result from the data

- ▣ Bitcoin 2014–2026 (weekly): GSADF 3.85 against a 95% critical value of 2.27: explosive episodes are detected
- ▣ One vs two Markov regimes for Bitcoin: $LR = 123.2$ against a parametric-bootstrap 95% critical value of 11.8 ($p = 0.005$)

Leads to

- ▣ Contagion from one market to the system (Chapter 18)

Chapter 18: Systemic Risk and Networks

Key idea

- Systemic risk: losses of one institution that spread to the whole system
- CoVaR (the VaR of the system conditional on the distress of one institution), SRISK and spillover networks measure the contribution of each institution

Key inference

- ΔCoVaR differences between banks: paired tests on joint block-bootstrap draws with a Holm correction; at $\alpha = 1\%$ only about 40 tail days, so extremal-quantile issues
- SRISK as published: LRMES simulated from GJR-GARCH + DCC [Brownlees & Engle \(2017\)](#); connectedness tables need bootstrap intervals and shrinkage VARs

Result from the data

- ΔCoVaR 1% of eleven banks: 4 of 25 within-region pairs differ at 5%, 0 after Holm; overlap of intervals alone would have ranked none
- Simulated LRMES (22 days, market below -10%): JPMorgan 10.6% today, 23.7% with the information of 31 March 2020

Leads to

- Stress testing and the team projects

Recap: Learning, Text and New Markets

- ▣ Flexible models need a strict process: purging, embargo, deflated performance measures
- ▣ Foundation models show no significant difference from strong volatility benchmarks and do not replace GARCH-based FHS for tail risk
- ▣ Text, crypto, bubbles and systemic risk reuse the same statistical toolkit
- ▣ New data and new models: the same five questions apply

Six Pitfalls That Returned in Every Chapter

Pitfalls of design

- ▣ **Look-ahead bias:** information from after t used at t
- ▣ **Overfitting:** a model tuned to the sample it is judged on
- ▣ **Multiple testing:** the best of many tries reported as if it were the only one

Pitfalls of data and inference

- ▣ **Wrong calendar join:** returns computed before aligning assets
- ▣ **Heavy tails ignored:** standard errors that assume the Normal distribution and independence
- ▣ **Wrong loss:** a loss function that cannot rank the forecast target

Pitfall 1: Look-Ahead Bias

- Any parameter, threshold or scaling estimated on the full sample leaks the future
 - ▶ Examples: full-sample GARCH parameters, full-sample standardisation, labels that overlap the test set
- BET, VaR 1%, 2005–2026: GARCH- t with full-sample parameters breaches on 1.31% of days, with rolling estimation on 1.47%
 - ▶ The look-ahead version looks better than any forecaster could have done in real time
- Chapter 13: shuffled K-fold reports 69.4% accuracy on pure noise
- Remedy: rolling or expanding windows; purging and embargo; write the information set $\mathcal{F}_{t-1}$ next to every forecast

Pitfall 2: Overfitting

- The in-sample fit rewards noise that will not repeat
 - ▶ Chapter 4: in-sample Sharpe 1.57, true Sharpe 0.30 for the same portfolios
- Backtest overfitting: trying many strategies and keeping the best [Bailey et al. \(2017\)](#)
 - ▶ The deflated Sharpe ratio corrects for the number of trials and for non-normal returns [Bailey & López de Prado \(2014\)](#)
- Remedy: fix the design before looking at the test period; report every model you tried; compare with a simple benchmark (1/N, historical simulation, random walk, log-HAR)

Pitfall 3: Multiple Testing

- Chapter 3: with 300 useless factors, on average 15.0 pass $|t| > 1.96$, and the best reaches $|t| \approx 3.06$
 - ▶ This is why the factor literature asks for $|t| > 3$ [Harvey, Liu & Zhu \(2016\)](#)
- Calendar anomalies, trading rules, model specifications: each extra test adds a chance of a false discovery
- Remedy: count your tests and correct
 - ▶ Bonferroni and Holm control the chance of any false discovery; Benjamini–Hochberg controls their share
 - ▶ For forecasts: the model confidence set instead of many pairwise tests [Hansen, Lunde & Nason \(2011\)](#)

Pitfall 4: Joining on Returns Instead of Prices

- ❑ Bitcoin trades every day, SPY only on weekdays
 - ▶ Wrong: compute each return on its own calendar, then keep the common days: Bitcoin's weekend moves disappear
 - ▶ Right: keep the common days of the **prices**, then compute returns: Monday covers Friday to Monday for both
- ❑ Bitcoin, 2020–2026, on the common days: cumulative return 1025% (right) against 680% (wrong)
 - ▶ Monday standard deviation 5.04% (right) against 3.83% (wrong): the wrong join hides weekend risk
- ❑ Remedy: align prices on common days first; one series alone stays on its own calendar

Pitfall 5: Heavy Tails Ignored in Inference

- Classical standard errors assume independence and finite fourth moments
 - ▶ BET losses: Hill tail index $\hat{\alpha} = 2.39$, block-bootstrap 95% CI [2.12, 3.03] [Hill \(1975\)](#): with $\alpha < 4$ the population kurtosis is infinite
 - ▶ So the sample kurtosis (11.1) estimates nothing, and a bootstrap interval for it is not valid [Athreya \(1987\)](#): report the tail index instead
- BET GARCH persistence: the Wald CI [0.974, 1.007] crosses the stationarity boundary $\alpha + \beta = 1$, where the Normal approximation fails [Andrews \(1999\)](#)
 - ▶ Profile-likelihood CI inside the parameter space $p \leq 1$: [0.975, 1] (IGARCH, $p = 1$, not rejected against the boundary critical value 2.71 efAndrewsB), so the half-life is at least 27 days and has no finite upper bound
- Remedy: robust (HAC) errors, block bootstrap where it is valid, tail indices instead of kurtosis, boundary-aware intervals

Pitfall 6: A Loss Function That Cannot Rank the Target

- Variance forecasts are compared with a noisy proxy (r_t^2 or RV_t)
 - ▶ With a conditionally unbiased proxy of the conditional variance (e.g. r_t^2 of a zero-mean return, or RV), MSE and QLIKE rank forecasts by expected loss as the true variance would; QLIKE depends only on the ratio proxy/forecast, so volatile days weigh less than under MSE [Patton \(2011\)](#)
 - ▶ Chapter 9: log-HAR beats GARCH in QLIKE; the MSE comparison is inconclusive
- ES alone has no strictly consistent scoring function on the usual broad classes of distributions; the pair (VaR, ES) has one, under mild conditions [Fissler & Ziegel \(2016\)](#)
 - ▶ Use FZ0 with Diebold–Mariano tests [Diebold & Mariano \(1995\)](#); expect finite-sample limits on precision [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)
- Remedy: choose the loss before the comparison, and state it in the project report

Recap: A Checklist Against the Pitfalls

Design

- ▣ Every forecast uses only $\mathcal{F}_{t-1}$
- ▣ Test period fixed in advance and used once
- ▣ All models tried are reported; tests are counted

Data and inference

- ▣ Prices joined before returns; adjustments and calendars documented
- ▣ Robust or bootstrap uncertainty for every headline number
- ▣ Loss function consistent for the target; comparison with a simple benchmark

One Question, Four Chapters

The question

- How much can a BET index position lose tomorrow, with probability 1%?
- Which forecast of that loss would we have trusted in real time, 2005–2026?

Data

- BET index, official daily closing values, 6 Jan 2000 – 18 Sep 2026: 6670 daily log returns
- Duplicate records (rows identical to the previous one) and weekend-dated rows removed; trading days with an unchanged close kept

The four steps

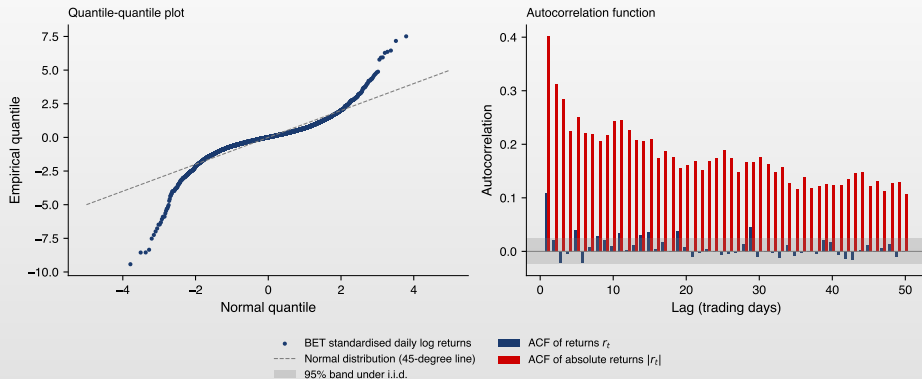
- Chapter 1: stylised facts, with robust inference
- Chapter 5: GARCH(1,1) with Student- t innovations
- Chapter 7: one-day VaR 1% from four models
- Chapter 8: coverage, independence and Basel zones



The Stock Exchange Palace, Bucharest (2023)

Photo: Chainwit. (2023); CC BY 4.0; Wikimedia Commons

Step 1: Stylised Facts of the BET

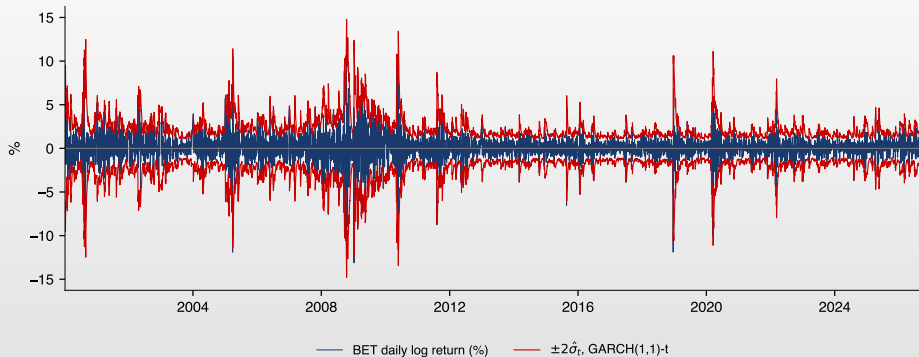


□ Left: quantiles of standardised returns against the Normal distribution; right: autocorrelations of r_t and $|r_t|$, lags 1–50

Reading Step 1

- Heavy tails: Hill index of losses $\hat{\alpha} = 2.39$ (block-bootstrap 95% CI [2.12, 3.03], $k = 153$); below 4, so kurtosis is not reported as an estimate
 - ▶ 51 days beyond ± 4 standard deviations, against 0.42 expected under the Normal distribution
 - ▶ Worst day: -13.1% (log) on 7 Jan 2009
- Clustering: first-order autocorrelation of $|r_t|$ is 0.40 and decays slowly; Ljung–Box $Q(10)$ on r_t^2 is 2440
 - ▶ Returns themselves have first-order autocorrelation 0.11: the BET is less liquid than large markets (Chapters 2, 10)
- Consequence: a Normal, constant-volatility VaR will be too low in stress and too high in calm periods

Step 2: GARCH(1,1)- t Volatility



- BET daily log returns with $\pm 2\hat{\sigma}_t$ from a GARCH(1,1) with Student- t innovations, estimated on 2000–2026

Reading Step 2

- ▣ Estimates (robust standard errors): $\hat{\alpha} = 0.192$ (0.025), $\hat{\beta} = 0.799$ (0.024), $\hat{\nu} = 5.01$
 - ▶ $\hat{\nu} \approx 5$: even after filtering, shocks have heavy tails
- ▣ Persistence $\hat{\alpha} + \hat{\beta} = 0.9907$: a half-life of about 74 trading days
 - ▶ Profile-likelihood 95% CI [0.975, 1]: persistence cannot be bounded away from 1 (Pitfall 5)
- ▣ Peak daily volatility 7.39% on 15 Oct 2008; at the end of the sample 1.67%

Step 3: Four VaR 1% Forecasts

Unconditional (last 500 days)

- ▣ **HS**: the loss $L = -r$ exceeded on 1% of the last 500 days (empirical quantile)
- ▣ **Normal**: $\text{VaR} = -(\hat{\mu} + \hat{\sigma} z_{0.01})$, $z_{0.01} = -2.326$

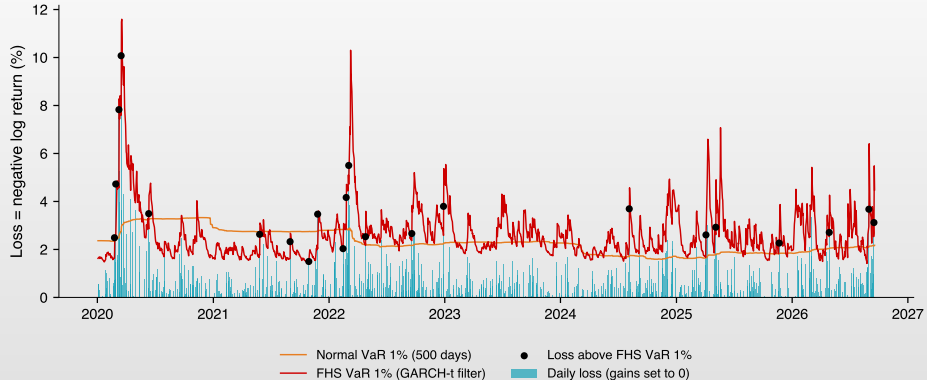
Conditional (GARCH- t on the last 1000 days)

- ▣ **GARCH- t** : $\text{VaR}_t = -(\hat{\mu} + \hat{\sigma}_t t_{\nu}^{-1}(0.01) \sqrt{(\nu - 2)/\nu})$
- ▣ **FHS**: $\text{VaR}_t = -(\hat{\mu} + \hat{\sigma}_t \hat{q}_{0.01}(z))$, the 1% quantile of the standardised residuals [McNeil & Frey \(2000\)](#)

Real-time design

- ▣ Forecast for day t uses data up to $t - 1$ only
- ▣ GARCH re-estimated every 21 trading days; $\hat{\sigma}_t$ updated daily
- ▣ Evaluation: 3 Jan 2005 – 18 Sep 2026, $T = 5\,432$ days
- ▣ Expected breaches at 1%: about 54

Step 3: Losses and VaR 1%, 2020–2026

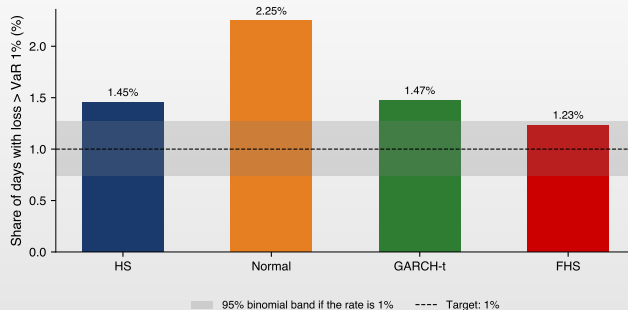


□ Daily BET losses (gains set to 0), Normal and FHS VaR 1%; black dots: losses above the FHS VaR

Reading Step 3

- FHS reacts within days: it peaked at 11.6% on 18 Mar 2020, during the COVID-19 crash
 - ▶ The Normal VaR on a 500-day window moves only slowly
- 2020–2026 (1 675 days, about 17 breaches expected): FHS 22 breaches, Normal 41
 - ▶ Normal breaches cluster in the first weeks of every stress episode
- Last forecast in the sample: FHS 4.48%, Normal 2.18%: the two models disagree most when volatility changes

Step 4: Breach Rates



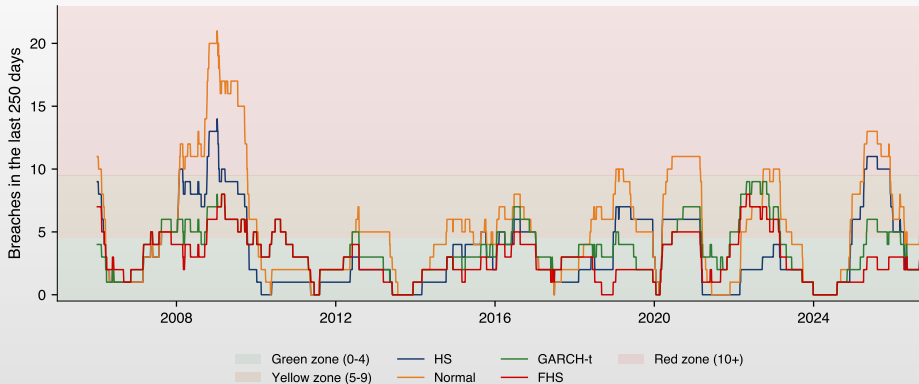
- Share of days with a loss above the VaR 1% forecast, 2005–2026; grey band: 95% binomial range if the true rate is 1%

Reading Step 4: Coverage and Independence

Model	Breaches	Rate	p_{uc}	p_{ind}	p_{cc}	Red zone
HS	79	1.45%	0.002	<0.001	<0.001	5.5%
Normal	122	2.25%	<0.001	<0.001	<0.001	18.6%
GARCH-t	80	1.47%	0.001	0.864	0.005	0.0%
FHS	67	1.23%	0.095	0.267	0.134	0.0%

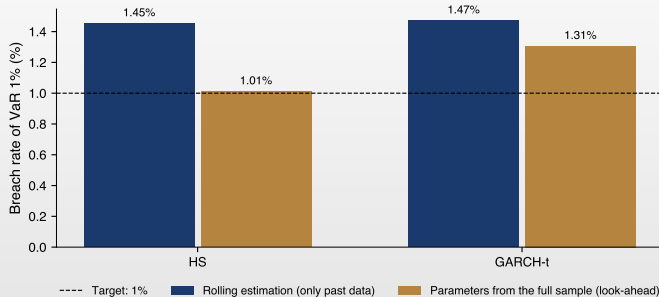
- ▣ p_{uc} : Kupiec coverage test; p_{ind} : Christoffersen independence test; p_{cc} : both jointly [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#)
- ▣ Red zone: share of 250-day windows with 10 or more breaches (Basel traffic light)
- ▣ Only FHS is rejected by none of the three tests at 5%; HS and Normal also fail independence: their breaches cluster
- ▣ GARCH-t: independence is not rejected, but there are too many breaches: the Student-t tail is not heavy enough at 1%

Step 4: Breaches in the Last 250 Days



- Number of breaches in a moving 250-day window, with the Basel green, yellow and red zones
- The Normal model spends 18.6% of the windows in the red zone, HS 5.5%, the two GARCH-filtered models never

The Look-Ahead Trap in the Case Study



- Full-sample HS quantile: 1.01% breaches, by construction; full-sample GARCH-t: 1.31%
- Both look better than their real-time versions: the gap is the value of information nobody had

Step 5: Comparing the Four Models Formally

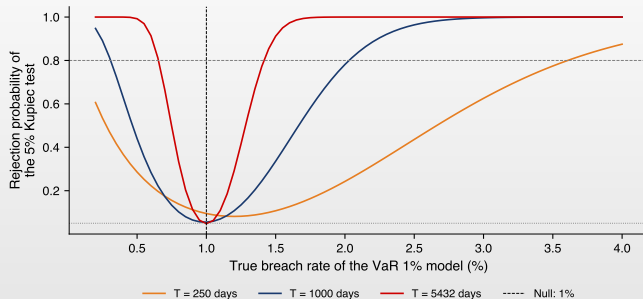
- **Dynamic Quantile** test: regress $I_t - 0.01$ on a constant, four lagged hits and VaR_t ; $DQ \sim \chi_6^2$ Engle & Manganelli (2004)
 - ▶ It detects breaches that are predictable from the past or from the level of VaR, which the Christoffersen test (one lag) can miss
- **FZ0 score** for the pair (VaR 2.5%, ES 2.5%), $v = -\text{VaR}$, $e = -\text{ES}$, return y : $S = -\frac{\mathbf{1}\{y \leq v\}(v-y)}{\alpha e} + \frac{v}{e} + \ln(-e) - 1$ Patton, Ziegel & Chen (2019)
 - ▶ Consistent for (VaR, ES) Fissler & Ziegel (2016); pairwise Diebold–Mariano tests with HAC variance Diebold & Mariano (1995); the model confidence set at 10% Hansen, Lunde & Nason (2011)
- **ES backtest**: comparative, against HS as the standard model; red if the model is significantly worse, green if significantly better, yellow otherwise Nolde & Ziegel (2017)
 - ▶ All forecasts use only data up to $t - 1$; 5 432 days, 3 Jan 2005 – 18 Sep 2026

Step 5: Results

Model	DQ	p_{DQ}	FZ0	p_{MCS}	ES zone
HS	192.1	<0.001	1.424	0.00	standard
Normal	364.6	<0.001	1.495	0.00	red
GARCH-t	17.9	0.007	1.175	0.18	green
FHS	9.9	0.129	1.157	1.00	green

- Only FHS passes the DQ test at 5% ($p = 0.129$); GARCH- t fails it ($p = 0.007$) through its excess of breaches
- FZ0: FHS and GARCH- t are not significantly different (DM $t = 1.39$), both beat HS and Normal ($|t| > 4.9$); the 10% model confidence set is {GARCH- t , FHS}
- Comparative ES backtest: Normal red, GARCH- t and FHS green: the ranking of Step 4 survives formal inference

How Much Can a Backtest Detect? The Power of Kupiec



- Exact binomial probability that the 5% Kupiec test rejects a VaR 1% model, as a function of its true breach rate, for $T = 250, 1000$ and 5432 days

Reading the Power Curves

- $T = 250$ (one Basel year): a true rate of 2% is detected only 24% of the time; rejection needs 0 or at least 7 breaches
 - ▶ Minimum detectable rate with 80% power: 3.65% at $T = 250$, 2.05% at $T = 1\,000$, 1.45% at $T = 5\,432$
- Our case study, $T = 5\,432$: a true rate of 1.3% would be detected with probability 55% only
- A non-rejection with low power is “no evidence”, not “no problem”: state the power before the test

Recap: The Case Study

- ▣ The BET shows every stylised fact: heavy tails, clustering, some linear autocorrelation
- ▣ GARCH- t captures clustering; the filtered residuals still have heavy tails
- ▣ A GARCH filter with an empirical tail (FHS) is the only VaR 1% that passes coverage, independence and the DQ test; with GARCH- t it forms the model confidence set for (VaR, ES)
- ▣ Real-time design matters: full-sample estimates flatter every model
- ▣ The same four steps are a template for your project

What the Project Is

- An empirical study of one question about financial markets, answered with the methods of this course
 - ▶ Teams; four steps: replicate, extend, deliver on GitHub, present and defend (next slide)
- It carries **70%** of the final grade; quizzes and seminar activity carry 20%, attendance 10%
- Starting points: the Part C question of every seminar (next section)
 - ▶ Your own question is welcome if it passes the five questions of this chapter
- A good project can end with a negative answer, if the evidence is honest and precise

The Project in Four Steps

- 1. **Replicate**: reproduce one full table or figure of a published paper (or of the lecture), cell by cell
 - ▶ Report the discrepancy of every cell and explain it; an AI assistant cannot guess a published table
- 2. **Extend**: new data, a new market (BVB, crypto) or a more recent sample, under a **pre-analysis plan** committed before the test period is opened
 - ▶ The plan states the null, the test, the loss, the sample split, the identification strategy, the power of the main test and the specification grid with its correction
- 3. **Deliver on GitHub**: code, lockfile, fixed seeds, data provenance; one notebook regenerates every number
 - ▶ `AI_USE.md` and `AI_ERRORS.md` are mandatory; the report is an 8–12-page paper
- 4. **Present and defend**: team presentation, then a 10-minute oral defence
 - ▶ Each member explains one result and answers a “what changes if...” question; grades can differ between members

Replication Standard

- Target: one full table or figure, with its DOI, page and table number, not a single coefficient
 - ▶ Report a cell-by-cell comparison: published value, your value, difference, and the reason for each gap (sample, filter, data vendor, definition)
- Large-scale replications show why: many anomalies shrink out of sample or with better data [Hou, Xue & Zhang \(2020\)](#), while most factors replicate once the design is right [Jensen, Kelly & Pedersen \(2023\)](#)
- Reproducibility: a lockfile of package versions, fixed random seeds, series names and dates of the data, one command that regenerates every number
 - ▶ Different teams on the same data disagree more than their standard errors suggest [Menkveld et al. \(2024\)](#): reproducible code narrows the gap

Identification: from Correlation to a Claim

- Event questions (“did spot ETFs, the FTSE upgrade or a bank tax change X?”) need a counterfactual
 - ▶ A control market not hit by the event: difference-in-differences on the risk measure
 - ▶ A synthetic control built from several untreated markets [Abadie, Diamond & Hainmueller \(2010\)](#); placebo event dates (e.g. $\pm$ one year)
- Unknown timing: a sup-Wald break test [Andrews \(1993\)](#); report the estimated break date, not an imposed one
- Forecasting questions: out of sample only, with a test period declared in the plan
- One event is $n = 1$: say what the design can and cannot identify

Power and Minimum Detectable Effect

- Before running the main test, compute the smallest effect the design can detect with 80% power at 5%
 - ▶ Kupiec on 250 days cannot distinguish a VaR 1% model from a 2% one: power 24%; minimum detectable rate 3.65% (Step 5)
- Deliverable: a simulated power curve of the main test at your T , under the null and under the alternative you care about
- Short samples of crypto, ETFs or post-event windows make this binding: many project questions are underpowered
- A non-rejection with low power is “no evidence”, not “no effect”

Robustness without p-Hacking

- The pre-analysis plan is a timestamped commit in the repository, made before the test period is opened
- Specification curve: estimate every defensible combination of window, lags, α and sample start; plot all estimates sorted, with their intervals [Simonsohn, Simmons & Nelson \(2020\)](#)
 - ▶ Report the spread across specifications, not the best cell
- Correct the family: Holm for any false discovery, Benjamini–Hochberg for their share, or a bootstrap reality check
 - ▶ False and missed discoveries in finance [Harvey & Liu \(2020\)](#); the scientific outlook of the field [Harvey \(2017\)](#)

Forecast Evaluation for Risk Projects

- Volatility: compare forecasts with a noisy, conditionally unbiased proxy only through a robust loss (MSE or QLIKE) [Patton \(2011\)](#)
 - ▶ Many models: the model confidence set [Hansen, Lunde & Nason \(2011\)](#), not the lowest average loss
- VaR: coverage, independence and the DQ test [Engle & Manganelli \(2004\)](#); with a benchmark, conditional predictive ability tests [Giacomini & White \(2006\)](#)
 - ▶ State the power of every backtest at your T
- ES: score the pair (VaR, ES) with FZ0; comparative backtests with three zones [Nolde & Ziegel \(2017\)](#); dynamic (VaR, ES) models [Patton, Ziegel & Chen \(2019\)](#)
 - ▶ The integrated case study (Steps 1–5) is the template

Choosing the Question

A good question is

- ▣ **Specific:** one market, one period, one quantity
- ▣ **Testable:** a null hypothesis and a benchmark
- ▣ **Feasible:** data you can obtain and clean in a few weeks
- ▣ **Relevant:** an answer matters to an investor, a bank or a regulator

Examples

- ▣ Too vague: "Is Bitcoin risky?"
 - ▶ Better: "Did spot ETFs change the one-day VaR 1% of Bitcoin, and which model forecasts it best since 2024?"
- ▣ Too broad: "Can machine learning predict the market?"
 - ▶ Better: "Does a boosting model beat log-HAR for next-day BET volatility, in QLIKE, 2018–2026?"

Data

- Use the course data where possible: daily series of indices, ETFs, BVB stocks, crypto and FX, plus intraday bars
 - ▶ Public sources: BNR reference rates, FRED, Kenneth French Data Library, DefiLlama; the BVB archive for Romanian indices
- Apply the course conventions
 - ▶ Adjusted prices for ETFs and stocks, closing values for indices, FX and crypto
 - ▶ Remove holiday fills and weekend quotes where the market is closed; annualise with the real frequency
 - ▶ Several assets: join on prices, then compute returns
- Look for bad prints before modelling: jumps of $\pm 50\%$ in adjusted prices are usually data errors

Methods: From Question to Chapter

If your question is about	Start from
Return distribution, tails, predictability	Chapters 1–2: moments, robust tests, variance ratios
Expected returns, factors, portfolios	Chapters 3–4: Fama–MacBeth, shrinkage, out-of-sample Sharpe tests
Volatility and correlation	Chapters 5, 6, 9: GARCH, DCC, copulas, log-HAR
Risk measures and capital	Chapters 7–8: FHS, EVT, backtests, FZ0
Liquidity, options, continuous time	Chapters 10–12: Amihud, Black–Scholes, VIX, variance risk premium
Forecasting with ML, deep learning or text	Chapters 13–15: purged CV, foundation models, sentiment
Crypto, bubbles, systemic risk	Chapters 16–18: stablecoins, GSADF, CoVaR, networks

Honest Evaluation Checklist

Before you look at the results

- ☐ Write the hypothesis, the benchmark and the loss function
- ☐ Fix the estimation window and the test period
- ☐ Decide how many models and parameters you will try, and the correction for them
- ☐ Write the identification strategy and the power of the main test

When you report

- ☐ Out-of-sample results only for claims about forecasting
- ☐ Every headline number with a standard error, CI or p -value
- ☐ Costs for trading strategies; subperiods and a second market for robustness
- ☐ What did **not** work, and why

The Report: an 8–12-Page Paper

- **1. Introduction:** question, hypothesis and contribution; **2. Literature:** 10–15 clickable references, at least 3 from the last five years, including [Franke, Härdle & Hafner \(2019\)](#), [McNeil, Frey & Embrechts \(2015\)](#)
- **3. Data and replication:** sources, period, cleaning rules; the replicated table cell by cell
- **4. Design:** the pre-analysis plan, identification strategy, power, benchmark, loss and tests
- **5. Results:** tables and charts with uncertainty; **6. Robustness:** the specification curve with the multiplicity correction; **7. Conclusion:** the answer in one sentence, and its limits
- **Code:** a GitHub repository whose README and notebook reproduce every number, with [AI_USE.md](#) and [AI_ERRORS.md](#)

The Presentation

Structure (about 12 minutes)

- ▣ The question and why it matters (1 slide)
- ▣ Data and one chart that shows the problem (2 slides)
- ▣ Method and benchmark (2 slides)
- ▣ Main result with its uncertainty (2–3 slides)
- ▣ Robustness, limitations, answer (2 slides)

Good practice

- ▣ One message per slide; charts with readable axes and the legend below
- ▣ Every team member presents and can answer questions on the code
- ▣ Then a 10-minute oral defence: each member explains one result and answers a “what changes if...” question, without AI

Project Grading (70% of the Final Grade)

Component	What we look for
Replication (step 1)	A published table or figure reproduced cell by cell; every gap explained
Question, data and extension (step 2)	Specific, testable question; documented data and calendar alignment
Methods and inference	Pre-analysis plan, identification strategy, power, real-time design, benchmark, specification curve with a multiplicity correction
Interpretation and report	Economic meaning, limitations, honest conclusion; an 8–12-page paper with 10–15 references
GitHub repository (step 3)	One command reproduces every result (lockfile, seeds); AI_USE.md, AI_ERRORS.md
Presentation and oral defence (step 4)	Individual: one result explained, answers to “what changes if...”
Total	70% of the final grade; plus quizzes and activity (20%) and attendance (10%)

- Grades can differ between team members, depending on the oral defence

Timeline

- Week 1 after this chapter: team and question
 - ▶ Submit a one-page proposal (template in Seminar 19, Part C); open the GitHub repository
- Weeks 2–3: data, the replication of one published table or figure (step 1) and the pre-analysis plan committed to the repository
- Weeks 4–5: extension, evaluation, robustness (step 2); feedback at the seminar
- Final weeks: repository with README and the two AI files (step 3); presentation and oral defence (step 4)
- Exact dates are announced at the course and seminar

Where Projects Usually Lose Points

- ▣ A question that changes after the results are known
- ▣ In-sample fit presented as forecasting power
- ▣ No benchmark, or a weak one chosen to be beaten
- ▣ Numbers without uncertainty; many tests without correction
- ▣ Charts without units, legends inside the data, unexplained acronyms
- ▣ Code that does not run, or numbers in the report that the code does not produce
- ▣ AI output kept without checks: undeclared use, unchecked code, references that do not exist

Recap: The Team Project

- ▣ One specific, testable question; one benchmark; one loss function chosen in advance
- ▣ Clean data with the course conventions; real-time design; honest reporting
- ▣ Replicate, extend, deliver on GitHub, present and defend: 70% of the grade

The AI Policy of the Course

- ▣ **Allowed:** AI assistants for code, debugging, literature search and writing, in the project and in the seminars
- ▣ **Declared:** every tool, what it was used for and what you checked yourselves, in [AI_USE.md](#)
 - ▶ Undeclared use of AI counts as plagiarism
- ▣ **Checked:** you are responsible for every line of code and text
 - ▶ “The AI wrote it” is not an explanation at the oral defence
 - ▶ Every reference needs a working DOI or link; a reference that does not exist counts as fabricated data
- ▣ **No AI during the oral defence:** you answer alone, from your own understanding of the project

The Two Mandatory Files

AI_USE.md: the declaration

- ▣ Tools: name, version, who used them and for what
- ▣ Files or sections where AI-written code or text was kept, and what you changed
- ▣ How you verified the replicated table, the data, the formulas and the references
- ▣ What the team did without AI

▣ Templates in English and Romanian: github.com/danpele/MFM/tree/main/project

AI_ERRORS.md: the error log

- ▣ At least **three** places where the assistant was wrong
- ▣ For each: what you asked, what it answered, what was wrong, how you found out, the correct version
- ▣ Errors you caught are a strength of the project, not a weakness

Typical AI Errors in Financial Modelling (1/2)

- ▣ **Look-ahead bias:** a rolling statistic or a scaler that includes day t in the forecast for day t
 - ▶ Check: change the last observation; the forecast for that day must not move; shift by one day (`.shift(1)`)
- ▣ **Sign of VaR:** the 1% quantile of returns reported as VaR, or the 99% quantile of returns (a gain) used as the loss
 - ▶ Check: $\text{VaR}_\alpha = -q_\alpha(r) > 0$ for a loss; one-day VaR 1% for an index is a few percent, not negative
- ▣ **Annualisation:** daily volatility multiplied by 252 instead of $\sqrt{252}$, or 252 days used for crypto that trades every day
 - ▶ Check: means scale with h , volatilities with $\sqrt{h}$; use the real number of trading days (365 for crypto)

Typical AI Errors in Financial Modelling (2/2)

- **Random cross-validation (CV) on time series:** shuffled K-fold trains on the future; in Chapter 13 it reports 69.4% accuracy on pure noise
 - ▶ Check: is any test date earlier than a training date? Use walk-forward or purged CV with an embargo [López de Prado \(2018\)](#)
- **Invented references:** plausible authors, title and journal, but the paper does not exist
 - ▶ Check: search every DOI on [Crossref](#); open the paper and find the claim you cite
- **Also frequent:** wrong degrees of freedom in a test, library defaults you did not choose, numbers stated without running any code
 - ▶ Every seminar has a C2 exercise that trains you to catch such errors

Recap: Using AI in Your Project

- Use AI freely, declare it in `AI_USE.md`, and check every line it writes
- The replicated table is your best test: an AI cannot guess it
- Log at least three AI errors in `AI_ERRORS.md`; at the oral defence you answer alone

Project Seeds: Part C of Every Seminar (1/2)

Ch.	Open question (link to the seminar)
0	Did spot ETFs change Bitcoin's correlation with equities?
1	Did BET-TR autocorrelation fall after the FTSE reclassification?
2	Did the BVB become more efficient after the 2020 FTSE upgrade?
3	Do factor premia exist on the Bucharest Stock Exchange?
4	An optimal portfolio of BVB blue chips in RON
5	Is the Romanian market's volatility driven by the US?
6	Did Romanian and euro-area equities become more integrated after 2020?
7	Capital for a BVB portfolio vs an S&P 500 portfolio

- Each seminar gives the data, a starting method and a reference analysis discussed in class

Project Seeds: Part C of Every Seminar (2/2)

Ch.	Open question (link to the seminar)
8	Which risk model would you certify for a Romanian bank's trading book?
9	Is Bitcoin's volatility more predictable than the S&P 500's?
10	Does illiquidity carry a premium on the BVB?
11	Which continuous-time model fits BET and Bitcoin?
12	Is the variance risk premium a tradable signal?
13	Does a foundation model beat boosting?
14	Can a zero-shot foundation model replace GARCH for the BET?
15	Does headline sentiment pay after costs?
16	Is Bitcoin an alternative asset, a risk asset or a hedge?
17	Is there an AI bubble?
18	How exposed are Romanian banks to euro-area banking stress?

- Chapter 20 (realised volatility with signatures, VOLARE data) also suits a forecasting project

Turning a Seed into a Project

- Replicate first: one published table or figure from a paper behind the seed (step 1)
- Narrow it: one market, one period, one number to forecast or test
 - ▶ Example: Seminar 8 becomes “Which of HS, FHS and GARCH-EVT gives a correct VaR 1% for a 50/50 BET–EUR portfolio in 2015–2026?”
- Extend it: a second market, a new model, a longer or more recent sample
 - ▶ Example: add a foundation model (Chapter 14) as a volatility filter inside FHS
- Keep the reference analysis as your benchmark and try to improve on it honestly

Research Question: How Much of a Result Is the Researcher?

- Menkveld et al. (2024), *Journal of Finance* 79(3): 164 research teams test the same six hypotheses on the same data (Finance Crowd Analysis Project, #fincap; Sec. I.A)
 - ▶ **Standard error**: uncertainty from sampling, the data-generating process; **nonstandard error**: uncertainty from the analysis path each researcher picks, the evidence-generating process (Introduction)
- **Model**: team j reports $X_j = e_j + \varepsilon_j$, with e_j the researcher fixed effect of its path and ε_j the sampling error (Sec. I.C, eq. 1)
 - ▶ Nonstandard error $:= Q_{0.75}(x) - Q_{0.25}(x)$, the interquartile range (IQR) of the estimates across teams (eq. 3); the IQR, not the standard deviation, because the e_j can have fat tails
- **Four questions**: how large is it; is it smaller for better and more reproducible work; does peer feedback reduce it; do researchers know its size? (Introduction)
 - ▶ Link to this chapter: every team project is one path through the “garden of forking paths”; pre-registration, the specification grid and a repository that reruns every number are the remedies

Data and Method, as Published

- **Data:** all trades in the Euro Stoxx 50 index futures, 2002–2018, 720 million records, each side flagged as client or own account (Sec. I.A)
 - ▶ Six hypotheses, each “no annual trend” in: market efficiency, realised bid–ask spread, client share of volume, realised spread on client orders, share of market orders among client orders, gross trading revenue of clients
 - ▶ Each team chooses its own measure and reports the average yearly change in % with its standard error (SE)
- **Four stages** in 2021: independent analysis (paper and code); feedback from two peer evaluators; the five best papers shared; final estimates without the code constraint (Sec. I.A)
 - ▶ A pre-analysis plan was filed publicly before the experiment started (Sec. I.A)
- **Quality:** team quality (first principal component of five proxies), reproducibility score 0–100 by the certification agency Cascad, demeaned peer rating (Sec. I.B)
 - ▶ Quantile regressions of the estimates at the 10th, 25th, 50th, 75th and 90th percentiles; a shrinking IQR means less dispersion (Sec. I.C)
- **Inference:** each estimate is tested against the median estimate, with a Bonferroni correction at a family level of 0.5% (Sec. I.C)
 - ▶ Multiverse analysis: the teams report their choice at each fork of Table V; the k -sample Anderson–Darling statistic ranks the forks by how much they move the estimates (Sec. II.C)

Key Results

- **Size:** for market efficiency the median estimate is -1.1% per year, with an IQR of 6.7 percentage points (pp); even for the client share of volume, a simple ratio, the IQR is 1.2 pp around -3.3% (Table I, Panel C)
 - ▶ Interpretation: the median SE of 2.5% implies a Gaussian IQR of $1.35 \times 2.5 = 3.4\%$, half the 6.7 pp; the no-dispersion null is rejected for all six hypotheses, $p < 0.0001$ (Table II)
- **Quality:** a rise of one standard deviation in reproducibility lowers the IQR by 25.0%, in peer rating by 33.3% (Table III, Fig. 2)
 - ▶ Interpretation: code that reruns end to end is associated with answers closer to the consensus
- **Feedback:** from stage 1 to stage 4 the IQR falls by 47.2% and the interdecile range by 68.2% (Table IV, Fig. 3)
 - ▶ Interpretation: peer review narrows the dispersion but does not remove it
- **Beliefs:** pooled over hypotheses, the teams' predicted standard deviation of estimates is 71.7% below the realised one (Sec. II.B.3, Fig. 4)
 - ▶ Interpretation: researchers are overconfident about how unique their own path is
- **Forks:** model choice and sampling frequency move the estimates most; a relative-change model gives a median annual trend of $-23\,000\%$ on daily data, -200% monthly and -4.56% yearly (Fig. 6, Fig. 7)
 - ▶ Interpretation: Jensen's inequality, $\prod_{t=1}^T \mathbb{E}(M_t) < \mathbb{E}(\prod_{t=1}^T M_t)$ (eq. 7); log differences and linear trends do not suffer from it

Replicate on Course Data, Limits, Extension

- **Replication:** a multiverse for the market-efficiency hypothesis on the Euro Stoxx 50 index and the BET (data/market), 2002–2018, then extended to the last full year (Seminar 19, C3)
 - ▶ Forks exactly as in Table V: outliers (none, winsorised or trimmed at the 2.5 and 97.5 percentiles), measure (variance ratio, R^2 of an AR model), frequencies (day to week, day to month), model (linear trend, log difference, relative difference)
 - ▶ Data limitation: daily closes only, so the intraday frequency pairs of Table V cannot be run
- **Limits stated by the authors:** the multiverse covers a few forks only and explains the reported estimates poorly (footnote 32); beliefs were asked as standard deviations, not IQRs (footnote 28)
 - ▶ The test of dispersion assumes consistent reported SEs (Sec. I.C); the dynamics of journal refereeing are kept out of scope (Sec. I.A)
- **Open extension:** does a pre-registered grid shrink the nonstandard error of a risk question? Measure the IQR of VaR 1% conclusions for the BET across defensible paths
 - ▶ For your project: report the distribution over your grid and the forks that move it most, not a single cell

An Open Question

How much of a result is the data, and how much the researcher?

- 164 teams tested the same hypotheses on the same data: the dispersion of their answers, the **nonstandard errors**, was sizable and underestimated by the teams [Menkveld et al. \(2024\)](#)
 - ▶ It was smaller for more reproducible research and fell after peer-review stages
- Replication is not hopeless: in a Bayesian model of replication most asset-pricing factors replicate and work out of sample in 93 countries [Jensen, Kelly & Pedersen \(2023\)](#)
- AI assistants make every extra specification almost free: the forking paths multiply, but so does the ability to report all of them
 - ▶ Formal version for a project: the conclusion must hold over a pre-registered set S of defensible specifications, with the family corrected (FWER or FDR); report its distribution over S , not the best cell

The Discovery Loop, Step by Step (1/2)

- **Week 1, the question:** sharpen a seed into H_0/H_1 with a replication target
 - ▶ “Here is our seed from Part C of Seminar N. Give three sharper, testable versions, the published table each would replicate and the series in data/market it needs.”
- **Weeks 1–2, literature and target:** the number to replicate is one you have seen in the PDF
 - ▶ “List the most cited and the most recent papers on this question with DOI; for each, the headline estimate, its table and its sample.”
 - ▶ Resolve every DOI on [Crossref](#); search with Semantic Scholar, Elicit or Google Scholar; open the table yourselves
- **Week 2, hypotheses and pre-registration,** committed before the first result
 - ▶ “For our H_0 and H_1 , list competing hypotheses and the implication that separates each; then draft a pre-registration: sample, filters, windows, tests, correction.”
 - ▶ The commit history of the repository is the evidence that the plan came first

The Discovery Loop, Step by Step (2/2)

- **Weeks 2–3, code and replication:** the replicated table becomes a test in the repository
 - ▶ `“Write a function that reproduces every cell of the published table from data/market and an assertion with the tolerance we accept.”`
- **Weeks 4–5, extension and robustness:** run the whole pre-registered grid
 - ▶ `“Here are all the cells we ran. Which conclusions survive a Holm correction, and which ones would a referee call data snooping?”`
- **Before the defence, critique:** the assistant asks, the team answers
 - ▶ `“Act as the examiner of our oral defence: ask the ten hardest questions on identification, inference and data. Do not answer them.”`
 - ▶ The team answers without AI, as at the defence itself
- **Report:** the distribution of results over the grid, not the best cell; prompts in AI_USE.md, caught errors in AI_ERRORS.md

What the Researcher Must Check

- **The replication target exists:** number, table and sample taken from the paper, not from the assistant's summary
 - ▶ Test: the report cites the page and the table; a second member finds the same number there
- **The plan came first**
 - ▶ Test: in the commit history, the pre-registration precedes the first results file
- **Forking paths:** more specifications tried than reported
 - ▶ Test: every run writes its configuration and result to a log; the report states their number and corrects the family
- **Reproducibility:** numbers entered manually in the report
 - ▶ Test: a fresh clone and one command regenerate every number to the reported digit
- **Understanding:** each member can derive the key formula and defend every choice
 - ▶ Test: an internal mock defence, without AI, one week before the real one

Mini-Case: Replicate, Then Extend

- **Replicate:** FHS VaR 1% for the BET; Chapter 8 reports 59 breaches in 4 937 days (1.20%, Kupiec $p = 0.181$)
 - ▶ The case study of this chapter: 67 in 5 432 days (1.23%, $p = 0.095$); same model, different evaluation start (2007 vs 2005) and settings: document every gap before calling a replication failed
- A typical assistant proposal, reading each p -value alone at 5%: “only GARCH-EVT is correctly calibrated on all four assets”
- **Extend:** the 24 conditional-coverage tests of Chapter 8 (4 assets $\times$ 6 models) form one family
 - ▶ Rejections at 5%: 12 without correction, 10 with Holm (FWER), 11 with BH (FDR 5%)
 - ▶ FHS fails only on the S&P 500 ($p = 0.040$); within the family of 24 it survives, so after Holm FHS and GARCH-EVT pass on all four assets
- Interpretation: “FHS fails for the S&P 500” depends on whether the test is read alone or in its family; pre-register the family and report both

Your Turn: the Capstone Plan

- **Question:** your seed from Part C, written as H_0/H_1 , with the observation that would falsify it
- **Replicate first:** one published table seen in the paper, turned into a passing test in the repository
 - ▶ Then extend over a pre-registered grid, with the family corrected and the full distribution reported
- **Deliverables** along the timeline: proposal and replication target, pre-registration commit, passing replication test, grid of results, mock defence, report, oral defence
- **AI log:** the prompts that changed the design in AI_USE.md; every caught error, with the test that caught it, in AI_ERRORS.md ([templates](#))

Key Takeaways

- ▣ Returns are hard to predict; their risk is not: volatility, tails and dependence can be forecast
- ▣ Every model is judged out of sample, against a simple benchmark, with a loss that fits the target
- ▣ Heavy tails change inference: use robust errors and the bootstrap, report intervals
- ▣ Most wrong results come from the design, not from the model: look-ahead, overfitting, multiple testing, calendars
- ▣ New tools (ML, foundation models, LLMs) raise the bar for evaluation, not lower it

Further Study

Textbooks

- Franke, Härdle & Hafner (2019); Härdle, Chen & Overbeck (2017); McNeil, Frey & Embrechts (2015); Tsay (2010)

Advanced courses on quantinar.com

-  Statistics of Financial Markets
-  Measuring Statistical Risk

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_19

References (1/5)

- ▣ Abadie, A., Diamond, A., Hainmueller, J. (2010). Synthetic control methods for comparative case studies: estimating the effect of California's tobacco control program. *Journal of the American Statistical Association*, 105(490), 493–505.
- ▣ Andrews, D.W.K. (1993). Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61(4), 821–856.
- ▣ Andrews, D.W.K. (1999). Estimation when a parameter is on a boundary. *Econometrica*, 67(6), 1341–1383.
- ▣ Athreya, K.B. (1987). Bootstrap of the mean in the infinite variance case. *Annals of Statistics*, 15(2), 724–731.
- ▣ Bailey, D.H., Borwein, J., López de Prado, M., Zhu, Q.J. (2017). The probability of backtest overfitting. *Journal of Computational Finance*, 20(4), 39–69.
- ▣ Bailey, D.H., López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- ▣ Barndorff-Nielsen, O.E., Shephard, N. (2006). Econometrics of testing for jumps in financial economics using bipower variation. *Journal of Financial Econometrics*, 4(1), 1–30.
- ▣ Black, F., Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
- ▣ Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- ▣ Bollerslev, T., Wooldridge, J.M. (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2), 143–172.
- ▣ Brownlees, C., Engle, R.F. (2017). SRISK: a conditional capital shortfall measure of systemic risk. *Review of Financial Studies*, 30(1), 48–79.
- ▣ Chow, K.V., Denning, K.C. (1993). A simple multiple variance ratio test. *Journal of Econometrics*, 58(3), 385–401.

References (2/5)

- ▣ Christoffersen, P.F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.
- ▣ Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- ▣ Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- ▣ DeMiguel, V., Garlappi, L., Uppal, R. (2009). Optimal versus naive diversification: how inefficient is the $1/N$ portfolio strategy?. *Review of Financial Studies*, 22(5), 1915–1953.
- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Engle, R. (2002). Dynamic conditional correlation: a simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3), 339–350.
- ▣ Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- ▣ Engle, R.F., Manganelli, S. (2004). CAViaR: conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- ▣ Fama, E.F. (1970). Efficient capital markets: a review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.
- ▣ Fissler, T., Ziegel, J.F. (2016). Higher order elicibility and Osband's principle. *Annals of Statistics*, 44(4), 1680–1707.
- ▣ Forbes, K.J., Rigobon, R. (2002). No contagion, only interdependence: measuring stock market comovements. *Journal of Finance*, 57(5), 2223–2261.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.

References (3/5)

- ▣ Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- ▣ Gibbons, M.R., Ross, S.A., Shanken, J. (1989). A test of the efficiency of a given portfolio. *Econometrica*, 57(5), 1121–1152.
- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). *Applied Quantitative Finance*, 3rd ed. Springer.
- ▣ Harvey, C.R. (2017). Presidential address: the scientific outlook in financial economics. *Journal of Finance*, 72(4), 1399–1440.
- ▣ Harvey, C.R., Liu, Y. (2020). False (and missed) discoveries in financial economics. *Journal of Finance*, 75(5), 2503–2553.
- ▣ Harvey, C.R., Liu, Y., Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- ▣ Hill, B.M. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5), 1163–1174.
- ▣ Hou, K., Xue, C., Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5), 2019–2133.
- ▣ Jensen, T.I., Kelly, B., Pedersen, L.H. (2023). Is there a replication crisis in finance?. *Journal of Finance*, 78(5), 2465–2518.
- ▣ Kupiec, P.H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2), 73–84.
- ▣ Ledoit, O., Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.

References (4/5)

- ▣ Lo, A.W. (2002). [The statistics of Sharpe ratios](#). *Financial Analysts Journal*, 58(4), 36–52.
- ▣ Lo, A.W., MacKinlay, A.C. (1988). [Stock market prices do not follow random walks: evidence from a simple specification test](#). *Review of Financial Studies*, 1(1), 41–66.
- ▣ Lobato, I., Nankervis, J.C., Savin, N.E. (2001). [Testing for autocorrelation using a modified Box–Pierce Q test](#). *International Economic Review*, 42(1), 187–205.
- ▣ López de Prado, M. (2018). [Advances in Financial Machine Learning](#). Wiley.
- ▣ Markowitz, H. (1952). [Portfolio selection](#). *Journal of Finance*, 7(1), 77–91.
- ▣ McNeil, A.J., Frey, R. (2000). [Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach](#). *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ McNeil, A.J., Frey, R., Embrechts, P. (2015). [Quantitative Risk Management: Concepts, Techniques and Tools](#), revised ed. Princeton University Press.
- ▣ Menkveld, A.J., Dreber, A., Holzmeister, F., Huber, J., Johannesson, M., Kirchler, M., et al. (2024). [Nonstandard errors](#). *Journal of Finance*, 79(3), 2339–2390.
- ▣ Nolde, N., Ziegel, J.F. (2017). [Elicitability and backtesting: perspectives for banking regulation](#). *Annals of Applied Statistics*, 11(4), 1833–1874.
- ▣ Patton, A.J. (2011). [Volatility forecast comparison using imperfect volatility proxies](#). *Journal of Econometrics*, 160(1), 246–256.
- ▣ Patton, A.J., Ziegel, J.F., Chen, R. (2019). [Dynamic semiparametric models for expected shortfall \(and value-at-risk\)](#). *Journal of Econometrics*, 211(2), 388–413.
- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). [Information entropy and measures of market risk](#). *Entropy*, 19(5), 226.

References (5/5)

- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). [Finite-sample precision limits for expected shortfall forecast comparisons](#). *Mathematics*, 14(13), 2316.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiat, M., Yatracos, Y.G. (2023). [Are cryptos becoming alternative assets?](#). *European Journal of Finance*, 29(10), 1064–1105.
- ▣ Phillips, P.C.B., Shi, S., Yu, J. (2015). [Testing for multiple bubbles: historical episodes of exuberance and collapse in the S&P 500](#). *International Economic Review*, 56(4), 1043–1078.
- ▣ Shanken, J. (1992). [On the estimation of beta-pricing models](#). *Review of Financial Studies*, 5(1), 1–33.
- ▣ Simonsohn, U., Simmons, J.P., Nelson, L.D. (2020). [Specification curve analysis](#). *Nature Human Behaviour*, 4(11), 1208–1214.
- ▣ Stambaugh, R.F. (1999). [Predictive regressions](#). *Journal of Financial Economics*, 54(3), 375–421.
- ▣ Tsay, R.S. (2010). [Analysis of Financial Time Series](#), 3rd ed. Wiley.
- ▣ Zhang, L., Mykland, P.A., Aït-Sahalia, Y. (2005). [A tale of two time scales: determining integrated volatility with noisy high-frequency data](#). *Journal of the American Statistical Association*, 100(472), 1394–1411.